Machine Learning System Design Interview
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Personalized News Feed

Introduction

A news feed is a feature of social network platforms that keeps users engaged by showing them recent activities that interest them. Most social networks such as Facebook, LinkedIn, Twitter, and Instagram (IG) personalize news feed to maintain user engagement.

Figure 10.1: Different types of news feed

In this chapter, we are asked to design a personalized news feed system.

Clarifying Requirements

Here is a typical interview exchange between a candidate and an interviewer.

Candidate: Can I assume the motivation for a personalized news feed is to keep users engaged with the platform?

Interviewer: Yes, we display sponsored ads between posts, and more engagement leads to increased revenue.

Candidate: Whose p user attributes (like friendship, like, and display posts with new activities to the user. Can I assume this activity feed is the same as the home page news feed on Facebook?

Interviewer: Yes it is an equivalent.

Candidate: Can a post contain textual content, images, videos, or any combination?

Interviewer: Yes, it can contain any of these.

Candidate: To keep users engaged, the system should place the most engaging content at the top of timeline, as people are more likely to read the top posts. Is this the goal, more engagement?

Interviewer: Yes, that is the goal.

Candidate: Is there a specific type of engagement we are optimizing for? I assume there are different types of engagement, such as clicks, likes, and comments. Different reactions have different values on our platform. For example, liking a post is more valuable than a click, and a comment is more valuable than a click and a like. Is this correct? Do we have value weights, (from Meta, "Feed ranks posts based on the engagement" and Meta paper on Ads click prediction)?

Candidate: What are the major interactions available on the platform? I assume users can click, like, share, comment, hide, block another user, and report inappropriate content. Are there other important interactions we should consider?

Interviewer: Yes, you can assume these are the major interactions.

Candidate: Do we need to consider the privacy settings of users when we retrieve content from timeline or open the application. If I take too long, users will get bored and leave. Let's assume the system should display the ranked posts in less than 200 milliseconds (ms).

Candidate: How many daily active users do we have? How many timeline updates do we expect each day?

Interviewer: Assume we have 1 billion daily active users (DAU), and each user on average visits the home page twice a day.

Let's summarize the clarification exchange. We are asked to design a personalized news feed system. The system retrieves ranked posts or posts with unseen comments, and ranks them based on how engaging they are to the user. This should take no longer than 200 ms. The objective of the system is to increase user engagement.

Frame the problem as an ML task

Defining the ML objective

Let's explore the following three possible ML objectives:

- Maximize the number of specific implicit interactions, such as dwell time or user clicks.
- Maximize the number of all types of implicit interactions.
- Maximize a weighted score based on both explicit and implicit interactions.

Let's discuss each option in more detail.

Option 1: Maximize the number of specific implicit interactions, such as dwell time or user clicks. In this option, our chosen metric signals as a proxy for user engagement. For example, we optimize for the system to maximize user clicks.

The advantage of this approach is that it is simple and easy to measure. The disadvantage is that it may not be the best proxy for user engagement.

The disadvantage is that implicit reactions do not always reflect a user's true opinion about a post. For example, a user may click on a post, but find it is not worth reading.

Option 2: Maximize the number of all implicit interactions, such as likes, shares, and clicks.

With this option, we maximize implicit reactions so people can spend more time on the platform.

The advantage of this approach is that implicit signals usually carry more weight than implicit signals. For example, a user liking a post is usually a more positive interaction than a click.

The disadvantage is that also like users actually express their opinion with explicit reactions. For example, a user may like a post engaging but not very useful. Also, it is very difficult to find the right mix between various other ranking points. Meta "Feed ranks posts based on the engagement" and Meta paper on ads click prediction.

Option 3: Maximize a weighted score based on both implicit and explicit reactions

Option 3 maximizes the number of specific implicit interactions that engaged users with a post. In particular, we assign a weight to each reaction, based on how strongly each reaction signals user engagement. For example, we assign the following weights to the different reactions:

- Click: 1
- Like: 2
- Comment: 10
- Share: 20
- Hide: -30
- Block: -50
- Report: -100

Meta paper states that "Feed ranks posts based on the engagement" and Meta paper on ads click prediction.

Figure 10.2: A personalized news feed system (new version)

Choosing the right ML category

A personalized news feed system produces a ranked list of posts based on how engaging the users are to a user. Ranking problems are ML problems. We can use a ranking algorithm to produce a ranked list of posts. Ranking algorithms are supervised learning algorithms. We can use a supervised learning algorithm such as binary classification or regression to rank posts.

Figure 10.3: Supervised learning algorithm

Since these probabilities are predicted, we compute the engagement score. Figure 10.4 shows an example of how the engagement score is calculated.

Reaction	Click	Like	Comment	Share	Hide	Block	Report
Weight	1	2	10	20	-30	-50	-100
Predicted probability	0.8	0.5	0.3	0.1	0.3	0.0001	0.0001
Value	0.8	1.0	3.0	2.0	-9.0	-0.005	-0.01
Score	0.31	2.9	1.2	3.8	-3.61	-0.0001	-0.00015

Engagement score = 0.00896

Figure 10.4: Calculate the engagement score

Data Preparation

Data engineering

A personalized news feed system also takes a variable list of posts based on how engaging the users are to a user. Professional networks such as LinkedIn, Twitter, and Instagram (IG) personalize news feed to maintain user engagement. Meta's "Feed ranks posts based on the engagement" and Meta's paper on ad click prediction discuss how they personalize their news feed.

As Figure 10.5 shows, we employ several library crawlers to predict the probabilities of various implicit and explicit reactions for a user post. user.

Reaction	Click	Like	Comment	Share	Hide	Block	Report
Predicted probability	0.8	0.5	0.3	0.1	0.3	0.0001	0.0001
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Reaction	Click	Like	Comment	Share	Friendship request	Hide	Block
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Engagement score = 0.00896

Figure 10.6: Calculate the engagement score

Data Preparation

Data engineering

Table 10.1: User data schema

Posts

Table 10.2 shows post data.

author_id	text	image_id	video_id	Timestamp
3	Today, is an fun video with my field trip.	IRL_id_pic45_sunset_sea	1ez5Dk	1656843334
7	Professional presentation	2Ft_id_pic_meeting	3ez5Dk	1656843335
245	Today I had a great experience. Used the to tell you about. I wish...		5ez5Dk	1656843340

Table 10.2: Post data

User-post interactions

Table 10.3 shows user-post interaction data.

User ID	Post ID	Interaction type	Interaction value	Location (lat, long)	Timestamp
9	78	Like	2	(40.7128, -74.0060)	1656843334
9	78	Share	20	(40.7128, -74.0060)	1656843335
11	78	Comment	Yes, look amazing	(34.0522, -118.2437)	1656843340
11	78	Block	-50	(34.0522, -118.2437)	1656843341
9	78	Impression		(37.3382, -121.8863)	1656843342

Table 10.3: User post interaction data

Friendships

The friendship table stores data of connections between users. We assume users can specify their close friends and family members. Table 10.4 shows examples of friendship data.

User ID 1	User ID 2	Time when friendship was formed	Close friend	Family member
15	1024	1656843332	False	False
11	26	1656843733	False	True
9	78	1656843334	True	False

Table 10.4: Friendship data

Feature engineering

In this article, we discuss extensive features and always check for the model. In particular, we engineer features from each of the following categories:

- Post features
- User features
- User-post features
- Friendship features
- Post-type features
- Image or video
- Text
- Hashtags
- Post's age

Post features

What is it? This is the textual content – the text body, of a post.

How to prepare it? Textual content helps determine what the post is about.

Why is it important? We can extract important signals from the post using natural language models to convert the text into a numerical vector. Since the textual content is usually in the form of sentences and not a single word, we use a context-aware language model such as BERT [4].

Image or video

What is it? A post may contain images or videos.

Why is it important? We can extract important signals from images. For example, an image of a pug may indicate that a post is cute for children.

How to prepare it? First, preprocess the images or videos. Next, use a pre-trained model to convert the unstructured megabytes data into a embedding vector. For example, we can use ResNet [5] or the recently introduced ECA model [6] as the pre-trained model.

Hashtags

What is it? This is the textual content – the text body, of a post.

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How to prepare it? These values are represented by numerical values. We scale these numerical values to bring them into a similar range.

Post's age

Why is it important? Users are more engaged to a group content around a certain topic. These hashtags represent the topics to which a post belongs. For example, a post with the hashtag #machinelearning, or #AI indicates that the post is related to technology. Hashtags in the model may decide to rank it higher for people who are interested in technology.

How to prepare it? First, preprocess the images or videos. Next, use a pre-trained model to convert the unstructured megabytes data into a embedding vector. For example, we can use ResNet [5], or the recently introduced ECA model [6] as the pre-trained model.

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**Reactions

Choosing the loss function

Multi-task models are trained to learn multiple tasks simultaneously. This means we need to compute the loss for each task separately and then combine them to get an overall loss. Typically, we define a loss function for each task depending on the ML category of the task. In our case, we use a binary cross-entropy loss for each binary classification task, and a regression loss such as MAE [11], MSE [12], or Huber loss [13] for the regression task (dwell-time prediction). The overall loss is computed by combining task-specific losses, as shown in Figure 10.13.

$$Loss = \lambda L_{dwell} + L_{skip} + L_{like} + \ldots + L_{share}$$

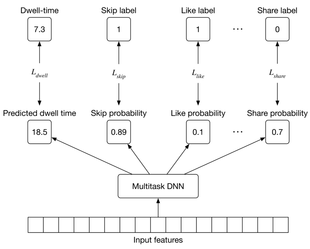

Figure 10.13: Training workflow

Evaluation

Offline metrics

During the offline evaluation, we measure the performance of our model in predicting different reactions. To evaluate the performance of an individual type of reaction, we can use binary classification metrics, such as precision and recall. However, these metrics alone may not be sufficient to understand the overall performance of a binary classification model. So, we use the ROC curve to understand the trade-off between the true positive rate and false positive rate. In addition, we compute the area under the ROC curve (ROC-AUC) to summarize the performance of the binary classification with a numerical value.

Online metrics

We use the following metrics to measure user engagement from various angles:

- Click-through rate (CTR)
- Reaction rate
- Total time spent
- User satisfaction rate found in a user survey

CTR. The ratio between the number of clicks and impressions.

$$CTR = \frac{\text{number of clicked posts}}{\text{number of impressions}}$$

A high CTR does not always indicate more user engagement. For example, users may click on a low-value clickbait post, and quickly realize it is not worth reading. Despite this limitation, it is an important metric to track.

Reaction rates. These are a set of metrics that reflect user reactions. For example, a like rate measures the ratio between posts liked and the total number of posts displayed in users' feeds.

$$\text{Like rate} = \frac{\text{number of liked posts}}{\text{number of impressions}}$$

Similarly, we track other reactions such as "share rate", "comment rate", "hide rate", "block rate", and "skip rate". These are stronger signals than CTR, as users have explicitly expressed a preference.

The metrics we discussed so far are based on user reactions. But what about passive users? These are users who tend not to react at all to the majority of posts. To capture the effectiveness of our personalized news feed system for passive users, we add the following two metrics.

Total time spent. This is the total time users spend on the timeline during a fixed period, such as 1 week. This metric measures the overall engagement of both passive and active users.

User satisfaction rate found in a user survey. Another way to measure the effectiveness of our personalized news feed system is to explicitly ask users for their opinion about the feed, or how engaging they find the posts. Since we seek explicit feedback, this is an accurate way to measure the system's effectiveness.

Serving

At serving time, the system serves requests by outputting a ranked list of posts. Figure 10.14 shows the architectural diagram of the personalized news feed system. The system comprises the following pipelines:

- Data preparation pipeline
- Prediction pipeline

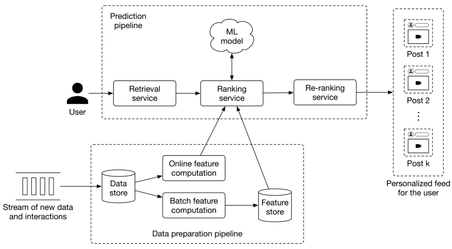

Figure 10.14: ML system design of a personalized news feed system

We do not go into detail about the data preparation pipeline because it is very similar to that described in Chapter 8, Ad Click Prediction in Social Platforms. Let's examine the prediction pipeline.

Prediction pipeline

The prediction pipeline consists of the following components: retrieval service, ranking service, and re-ranking service.

Retrieval service

This component retrieves posts that a user has not seen, or which have comments also unseen by them. To learn more about efficiently fetching unseen posts, read [14].

Ranking service

This component ranks the retrieved posts by assigning an engagement score to each one.

Re-ranking service

This service modifies the list of posts by incorporating additional logic and user filters. For example, if a user has explicitly expressed interest in a certain topic, such as soccer, this service assigns a higher rank to the post.

Other Talking Points

If there is time left at the end of the interview, here are some additional talking points:

- How to handle posts that are going viral [15].
- How to personalize the news feed for new users [16].
- How to mitigate the positional bias present in the system [17].
- How to determine a proper retraining frequency [18].

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